Chapter 9

Biostatistics and Data Analysis

Contents

- Chances of error
- Types of errors
- Level of Significance
- Confidence Levels
- P value
- Power of a test

CHANCES OF ERROR

Definition

- Chances of error refer to the probability of making incorrect conclusions or incorrect decisions in statistical testing.

TWO TYPES OF ERRORS

1. Type I Error
2. Type II Error

TWO TYPES OF ERRORS

1. Type I Error

- A Type I error is also known as a "false positive error“ or “α Error”.
- Rejecting the null hypothesis ($\mathbf{H_0}$) when it is true.

2. Type II Error

- A Type II error is also known as a "false negative error“ or “β Error”.
- Failing to reject **the null hypothesis ($\mathbf{H_0}$) when it is false.**

Test Result	**Truth**	
	H_0 Correct	**H_0 wrong**
Reject H_0	Type I or α Error	No error
Failing to reject H_0	No error	Type II or β Error

TWO TYPES OF ERRORS

For example:

H_0: *The medicine has effect / does work*

Test Result	**Truth**	
	H_0 Correct: it does work.	**H_0 wrong:** it actually doesn't work.
Reject H_0	**Type I Error** **Test Result:** Conclude medicine doesn't work	**No error** **Test Result:** Conclude medicine doesn't work.
Fail to Reject H_0	**No error** **Test Result:** Conclude medicine does work.	**Type II Error** **Test Result:** Conclude medicine does work

TWO TYPES OF ERRORS

For example:

H_0: *The medicine has no effect / does not work*

Test Result	**Truth**	
	H_0 Correct: it doesn't work.	**H_0 wrong:** it actually does work.
Reject H_0	**Type I Error** **Test Result:** Conclude medicine does work.	**No error** **Test Result:** Conclude medicine does work.
Fail to Reject H_0	**No error** **Test Result:** Conclude medicine doesn't work	**Type II Error** **Test Result:** Conclude medicine doesn't work

TWO TYPES OF ERRORS

➢ In 1980s, 20–29 year old men in Dawadmi had a mean (μ) body weight of 170 pounds (77kg). Standard deviation σ was 40 pounds. We test whether mean body weight in the population now differs.

✓**Research Question:** Has the mean body weight of 20-29 year old men in Dawadmi changed since the 1980s?

✓**Null hypothesis** $H_{0:}$

❖$\mu = 170$ ("no difference")

➢ **One-Sided Alternative Hypothesis (Increased): $\mu > 170$**

✓ **Type I Error (α-error):** Concluding $\mu > 170$ when $\mu = 170$ (False Positive)

❖ Example: Study finds mean weight increased to 180 pounds, but it's actually still 170 pounds.

✓ **Type II Error (β-error):** Failing to detect $\mu > 170$ when $\mu > 170$ (False Negative)

❖ Example: Study fails to detect a real increase in mean weight from 170 to 180 pounds.

TWO TYPES OF ERRORS

➢ In 1980s, 20–29 year old men in Dawadmi had a mean (μ) body weight of 170 pounds (77kg). Standard deviation σ was 40 pounds. We test whether mean body weight in the population now differs.

✓**Research Question:** Has the mean body weight of 20-29 year old men in Dawadmi changed since the 1980s?

✓**Null hypothesis** $H_{0:}$

❖$\mu = 170$ ("no difference")

➢ **One-Sided Alternative Hypothesis (Decreased): $\mu < 170$**

✓ **Type I Error (α-error):** Concluding $\mu < 170$ when $\mu = 170$ (False Positive)

❖ Example: Study finds mean weight decreased to 160 pounds, but it's actually still 170 pounds.

✓ **Type II Error (β-error):** Failing to detect $\mu < 170$ when $\mu < 170$ (False Negative)

❖ Example: Study fails to detect a real decrease in mean weight from 170 to 160 pounds.

TWO TYPES OF ERRORS

➢ In 1980s, 20–29 year old men in Dawadmi had a mean (μ) body weight of 170 pounds (77kg). Standard deviation σ was 40 pounds. We test whether mean body weight in the population now differs.

✓**Research Question:** Has the mean body weight of 20-29 year old men in Dawadmi changed since the 1980s?

✓**Null hypothesis** $H_{0:}$

❖$\mu = 170$ ("no difference")

➢ **Two-Sided Alternative Hypothesis: $\mu \neq 170$**

✓ **Type I Error (α-error):** Concluding $\mu \neq 170$ when $\mu = 170$ (False Positive)

❖ Example: Study finds mean weight changed (increased or decreased), but it's actually still 170 pounds.

✓ **Type II Error (β-error):** Failing to detect $\mu \neq 170$ when $\mu \neq 170$ (False Negative)

❖ Example: Study fails to detect a real change in mean weight (increase or decrease) from 170 pounds.

LEVEL OF SIGNIFICANCE (α)

Definition:

- It is the probability of making **Type I error**.

OR

- The probability of rejecting the null hypothesis H_0 when it is actually true.

Key Points:

- Denoted by: **α**
- α is typically set before conducting a hypothesis test. (TRUE)
- Related to errors: $P(Type\ I\ Error) = \alpha$
  - ✓ Controls the risk of making a **false positive** error (*Type I Error*).

Example:

If **α = 0.05**

- 5% chance of rejecting H_0 when it's true (5% chance of Type I Error)
- 95% confidence in the decision (95% chance of correct decisions)

LEVEL OF SIGNIFICANCE (α)

How to Calculate Significance Level (α) ✗

- It is **NOT calculated from data.**
- α is chosen BEFORE doing the test.
- you *select* it based on how strict you want your test to be.

Common choices:

- $\alpha = 0.05$
- $\alpha = 0.01$
- $\alpha = 0.001$

How to choose α?

- If you want *moderate* strictness → choose **0.05**
- If you want *high* strictness → choose **0.01**
- If you want *extremely high* strictness → choose **0.001**

👉 **So α is normally not calculated — it is selected.**

CONFIDENCE LEVELS

Definition

- A confidence level is the percentage that reflects how sure you are about your estimate.

OR

- Confidence Level = Confidence in correct decisions.

Confidence Level Relationship to Significance Level (α):

- Confidence $Level = 1 - \alpha$
  - ✓ If **α** is the probability of making a Type I error (***P(Type I Error)***), the confidence level represents how confident you are in your statistical decision.

CONFIDENCE LEVELS

Confidence Levels

1. 95% confidence level ($\alpha = 0.05$) "Significant"
2. 99% confidence level ($\alpha = 0.01$) "Highly Significant"
3. 99.9% confidence level ($\alpha = 0.001$) "Very Highly Significant"

Interpretation

1. **Significant (95%):** Strong evidence against null hypothesis.
2. **Highly Significant (99%):** Very strong evidence against null hypothesis.
3. **Very Highly Significant (99.9%):** Extremely strong evidence against null hypothesis.

CONFIDENCE LEVELS

How to Calculate Confidence Level

- Once you choose α, the **confidence level** is automatically determined using:

$$\text{Confidence } Level = 1 - \alpha$$

Examples:

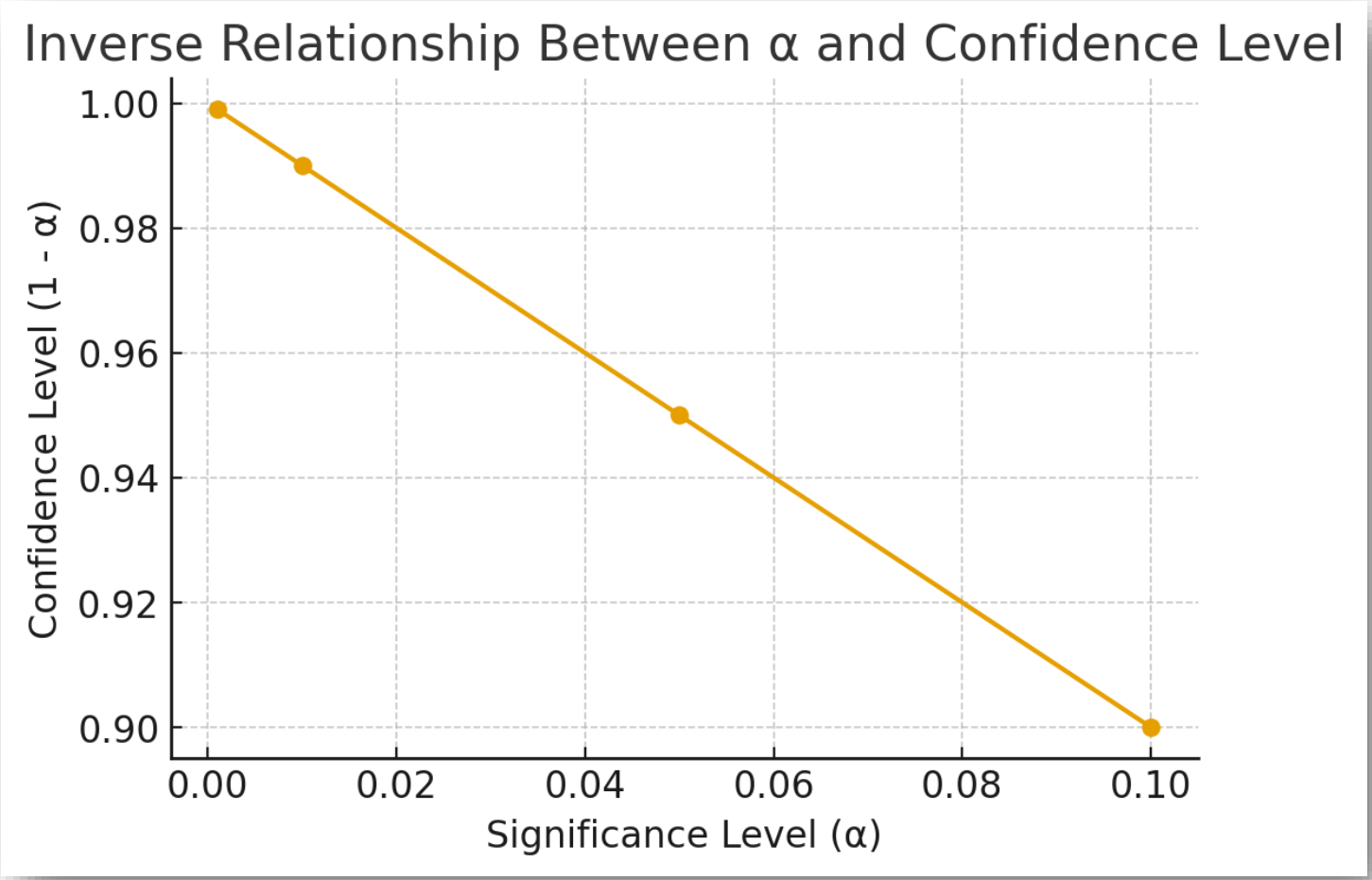
α	Confidence Level
0.05	1 − 0.05 = **0.95 (95%)**
0.01	1 − 0.01 = **0.99 (99%)**
0.001	1 − 0.001 = **0.999 (99.9%)**

Inverse Relationship Between α and Confidence Level

- This means:
  - ✓ When **α increases**, confidence level **decreases**
  - ✓ When **α decreases**, confidence level **increases**

CONFIDENCE LEVELS

Confidence Levels & Their Significance



CONFIDENCE LEVELS

Confidence Levels & Their Significance

Confidence Level	Significance Level (α)	Term Used	Meaning
95%	0.05	Significant	➢ Moderate evidence against H_0; ➢ 5% chance of Type I error
99%	0.01	Highly Significant	➢ Strong evidence against H_0; ➢ only 1% chance of Type I error
99.9%	0.001	Very Highly Significant	➢ Extremely strong evidence against H_0; ➢ only 0.1% chance of Type I error

Simple One-Line Explanation:

- Lower confidence = Higher α = More risk of Type I error
- Higher confidence = Lower α = Less risk of Type I error

POWER OF A TEST

Definition

- The probability of rejecting the null hypothesis (H0) when it is false.
- Power of a test can be written mathematically as:
  - ✓ Power=1− P(Type II error) (Whereas; β = P(Type II error))
  - ✓ Power $= 1 - \beta$

P VALUE

- The p-value, or probability value, measures the strength of evidence against a null hypothesis in statistical hypothesis testing.

Interpretation

- Small p-value (e.g., 0.01): Strong evidence against null hypothesis.
- Large p-value (e.g., 0.50): Weak evidence against null hypothesis.
- If the p-value is less than the significance level (α), Reject the null hypothesis.

Interpretation Guidelines for P-values

P-value < 0.01	➤ Strong evidence against H_0 ➤ Result is highly significant
$0.01 \leq$ P-value < 0.05	➤ Moderate evidence against H_0 ➤ Result is significant
$0.05 \leq$ P-value < 0.10	➤ Weak evidence against H_0 ➤ Result is marginally significant
P-value ≥ 0.10	➤ No evidence against H_0 ➤ Result is not significant

Practice for students

"Designed for students to practice and gain familiarity with different types of statistical problems and cases, enhancing their understanding of key concepts", like:

- How to Calculate Test Statistic (z-score)
- How to Calculate p-value
- Set significance level (α)
- How to Make Interpretation

Test for population mean

Case I (Large sample)

- The daily yield at a chemical plant, recorded for 50 days, possesses a sample mean of 884 tons. The population s.d is known to be 42 tons. Do the data provide enough evidence to indicate that the average daily yield of the chemical plant is 900 tons per day. Perform a test the data support the claim, using p-value approach.

Test for population mean

Case I (Large sample)

Step 1: Define the Problem and Hypotheses

- Research Question: Is the average daily yield of the chemical plant 900 tons per day?
- Null Hypothesis (H0): $\mu = 900$ (average daily yield is 900 tons)
- Alternative Hypothesis (H1): $\mu \neq 900$ (average daily yield is not 900 tons)

Step 2: Gather Data and Calculate Sample Statistics

- Sample size (n) = 50 days
- Sample mean ($\bar{x}$) = 884 tons
- Population standard deviation (σ) = 42 tons

Step 3: Calculate Test Statistic (z-score)

- Formula: $z = (\bar{x} - \mu 0) / (\sigma / \sqrt{n})$
- Calculation: $z = (884 - 900) / (42 / \sqrt{50}) = -2.69$

Test for population mean

Case I (Large sample)

Step 4: Calculate p-value

- p-value = $2P(Z > |z0|) = 2P(Z > 2.69)$
- Using z-table or calculator: p-value = 2 x 0.0036 = 0.0072

Step 5: Make Decision and Interpret Results

- Compare p-value to significance level ($\alpha = 0.05$)
- Since p-value (0.0072) < α (0.05), reject H0
- Conclusion: Data provide enough evidence to conclude that the mean daily yield differs from 900 tons.

Interpretation:

- The rejection of H0 implies that the average daily yield of the chemical plant is unlikely to be 900 tons. The actual yield may be higher or lower than 900 tons.

Test for population mean

Case II (Large sample)

- The average nicotine content of a particular brand of cigarette is supposed to be **1.5 mg per cigarette**. A smoker of this brand wants to see if it is really **1.5 mg or more**. With this purpose of the nicotine contents of **40 cigarettes** are measured. The sample **mean based on 40 cigarettes is 1.62 mg** per cigarette. The population **S.D is 0.5 mg.** Do the data provide enough evidence that the average **nicotine contents exceeds 1.5** mg? **Test** using **p-value approach.**

Test for population mean

Case II (Large sample)

Step 1: Define Hypotheses

- Research Question: Is the average nicotine content of the cigarettes more than 1.5 mg?
- Null Hypothesis (H0): $\mu = 1.5$ mg (average nicotine content is 1.5 mg per cigarette)
- Alternative Hypothesis (H1): $\mu > 1.5$ mg (average nicotine content exceeds 1.5 mg per cigarette)

Step 2: Gather Data

- Sample size (n) = 40 cigarette
- Sample mean ($\bar{x}$) = 1.62 mg nicotine per cigarette
- Population standard deviation (σ) = 0.5 mg

Step 3: Calculate Test Statistic (z-score)

$$z = (\bar{x} - \mu 0) / (\sigma / \sqrt{n})$$

$$= (1.62 - 1.5) / (0.5 / \sqrt{40})$$

$$= 1.52$$

Test for population mean

Case II (Large sample)

Step 4: Calculate p-value

p-value = $P(Z > |z0|) = P(Z > 1.52)$

Using z-table or calculator: p-value = 0.0645

Step 5: Make Decision

- Compare p-value to significance level ($\alpha = 0.05$)
- Since p-value (0.0645) > α (0.05), fail to reject H0.
- Conclusion: Data do not provide enough evidence to conclude that the average nicotine content exceeds 1.5 mg.

Interpretation

- The sample mean (1.62 mg) suggests slightly higher nicotine content, but the evidence is not strong enough to reject the null hypothesis.


ThankYou
Adobe Stock | #271928289